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Public Health Statistics: Continuous Data Measures

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Continuous Data: Useful Summary Statistics

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Learning Objectives

- ▶ Upon completion of this lecture, you will be able to:
 - ▶ Compute a sample mean and standard deviation
 - ▶ Interpret the estimated mean, standard deviation, median, and various percentiles computed for a sample of continuous data measures

Summarizing and Describing Continuous Data

- ▶ Measures of the center of data
 - ▶ Mean
 - ▶ Median (50th percentile)
- ▶ Measure of data variability
 - ▶ Standard deviation
- ▶ Other measures of location
 - ▶ Percentiles

Sample Mean: The Average or Arithmetic Mean

- ▶ Add up data, then divide by sample size (n)
- ▶ The sample size n is the number of observations (pieces of data)
- ▶ Example: mean, small systolic blood pressure (SBP) dataset

Example: Mean, Small SBP Dataset—1

- ▶ Five systolic blood pressures (mmHg), $n=5$: 120 mmHg, 80 mmHg, 90 mmHg, 110 mmHg, 95 mmHg
- ▶ Can be represented with math type notation: $x_1=120, x_2=80, \dots, x_5=95$
- ▶ The sample mean is computed by adding up the five values and dividing by 5
 - ▶ In stat notation, the sample mean is frequently represented by a letter with a line over it
 - ▶ Example: $\bar{x}$ (pronounced “x bar”)

Example: Mean, Small SBP Dataset—2

- ▶ Five systolic blood pressures (mmHg), $n=5$: 120 mmHg, 80 mmHg, 90 mmHg, 110 mmHg, 95 mmHg
- ▶ Can be represented with math type notation: $x_1=120, x_2=80, \dots, x_5=95$
- ▶ The sample mean:

$$\bar{x} = \frac{120 + 80 + 90 + 110 + 95}{5} = 99 \text{ mmHg}$$

The Sample Mean, Generally Speaking—1

- ▶ Generic formula representation

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}, \text{ where } \sum_{i=1}^n x_i = x_1 + x_2 + x_3 + \dots \dots x_n$$

- ▶ In the formula, to find the mean, we use the “summation sign,” Σ
 - ▶ This is just mathematical shorthand for “add up all of the observations”

The Sample Mean, Generally Speaking—2

- ▶ Also called *sample average* or *arithmetic mean*
- ▶ Why is it called the *sample* mean?
 - ▶ To distinguish it from population mean (an unknown, unknowable value of interest μ , that can be estimated by $\bar{x}$)
- ▶ Sensitive to extreme values (in smaller samples)
 - ▶ A change in the value of one data point could make a substantial change in the values of a sample mean

Example: The Median, Small SBP Data Set—1

- ▶ The median is the middle value in an ordered set of continuous data measures (the median is also called the 50th percentile)
- ▶ The median value of the five SBP measurements is 95 mmHg:

80 90 95 110 120



Example: The Median, Small SBP Data Set—2

- ▶ The sample median is not sensitive to the influence of extreme sample values (unlike the sample mean)
 - ▶ For example, in the sample of five SBP measurements, if the value 120 was changed to 200, the sample median would remain the same, but the *sample mean would increase from 99 mmHg to 115 mmHg*

80 90 95 110 200



The sample median is still 95 mmHg

Example: The Median, Small SBP Data Set—3

- ▶ If the sample size is an even number, then the median is the average of two middle values
- ▶ Suppose we add a sixth SBP value to the original SBP values. The (now) six values, in ascending order:

80 90 95 110 120 125



The sample median of these six values is $\frac{95+110}{2} = 102.5$ mmHg

Describing Variability: The Sample Variance

- ▶ Sample variance (s^2)
- ▶ Sample standard deviation (s or SD)
- ▶ The sample variance is the average of the square of the deviations about the sample mean

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

Describing Variability: Sample Standard Deviation (s)

- ▶ The sample standard deviation is the square root of the sample variance, s^2

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

Example: Standard Deviation, Small SBP Data Set—1

- ▶ Systolic blood pressures (mmHg), $n=5$: 120 mmHg, 80 mmHg, 90 mmHg, 110 mmHg, 95 mmHg. The mean, $\bar{x}$, is 99 mmHg.
- ▶ The sample variance computation, numerator:

$$\begin{aligned}\sum_{i=1}^5 (x_i - \bar{x})^2 &= \\ \sum_{i=1}^5 (x_i - 99)^2 &= 21^2 + (-19)^2 + (-9)^2 + 11^2 + \\ &\quad (-4)^2 = 1,020 \text{ mmHg}^2\end{aligned}$$

Example: Standard Deviation, Small SBP Data Set—2

- ▶ Sample variance computation, complete

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} = \frac{1,020}{4} = 255 \text{ mmHg}^2$$

- ▶ Sample standard deviation (s)

$$s = \sqrt{s^2} = \sqrt{255 \text{ mmHg}} \approx 15.97 \text{ mmHg} = 16 \text{ mmHg}$$

The Sample Standard Deviation (s), Generally Speaking—1

- ▶ The more variability there is in a sample of data, the larger the value of s
- ▶ s measures the variability (spread) of the individual sample values around the sample mean
- ▶ s can equal 0 only if there is no variability (if all n sample observations have the same value)
- ▶ The units of s are the same as the units of the data measurements in the sample (for example, mmHg)
- ▶ Often abbreviated SD or sd
- ▶ s^2 is the best estimate from the sample of the population variance σ^2 ; s is the best estimate of the population standard deviation σ

The Sample Standard Deviation (s), Generally Speaking—2

- ▶ In the formula for s , the denominator is $n-1$

$$s^2 = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

- ▶ Why divide by $n-1$ instead of n ?

Example: Standard Deviation, Larger SBP Data Set—1

- ▶ SBP measurements taken for a sample of 113 men ($n=113$)
- ▶ The first 50 sample values are shown here:

142	116	137	126	124
123	116	127	115	129
107	103	130	133	116
129	117	131	107	138
114	113	142	120	147
105	122	111	111	129
132	89	134	121	120
128	120	119	112	139
121	124	132	140	120
116	152	123	131	141

Example: Standard Deviation, Larger SBP Data Set—2

- ▶ This, of course, is too much data ($n=113$) to compute summary statistics by hand
- ▶ The results from a computer are as follows:

(Estimate of μ): $\bar{x} = 123.6$ mmHg

(Estimate of σ): $s = 12.9$ mmHg

(Estimate of population median, m): $\hat{m} = 123.0$ mmHg

Percentiles in Samples With All Unique Values

- ▶ Other values that can help us quantify the distribution of continuous data values include the sample percentiles (as estimates of the underlying population percentiles)
- ▶ In general, if all sample values are unique, the p^{th} sample percentile is that value in a sample of data such that p percent of the sample values are less than or equal to this value, and $(100-p)$ percent are greater than this value (example: the median is the 50th percentile)
- ▶ Percentiles can be computed by hand but are generally done via computer

Percentiles in Samples with Some Repeated Values

- ▶ If not all sample values are unique, i.e. some value are repeated, then the p^{th} sample percentile is that value in a sample of data such that p percent of the sample values are less than or equal to this value, and $(100-p)$ percent are greater than (or equal to) this value (example: the median is the 50th percentile)
- ▶ In such samples, some percentiles may have same values

Example: Percentiles, Larger SBP Data Set—1

- ▶ Systolic blood pressure (SBP) measurements from a random sample of 113 adult men taken from a clinical population (based on results from a computer)
- ▶ The 10^{th} *percentile* for these 113 blood pressure measurements is 107 mmHg, meaning that approximately 10% of the men in the sample have $SBP \leq 107$ mmHg, and $(100-10) = 90\%$ of the men have $SBP > 107$ mmHg
- ▶ The 75^{th} *percentile* for these 113 blood pressure measurements is 132 mmHg, meaning that approximately 75% of the men in the sample have $SBP \leq 132$ mmHg, and $(100-75) = 25\%$ of the men have $SBP > 132$ mmHg

Example: Percentiles, Larger SBP Data Set—2

- ▶ SBP measurements from a random sample of 113 adult men taken from a clinical population (based on results from a computer)
- ▶ Some percentiles of these data, based on the results from a computer:

Percentile	Sample estimate
2.5 th	100.7 mmHg
25 th	114 mmHg
50 th	123 mmHg
75 th	132 mmHg
97.5 th	151.2 mmHg

Example: Heritage Health Length of Stay Data—1

- ▶ Length of stay claims at Heritage Health, with an inpatient stay of at least one day, in 2011¹ (12,928 claims)
- ▶ The first 50 measurements:

21	1	1	1	1
1	4	3	2	7
12	1	2	2	7
5	8	3	1	1
1	2	6	1	1
1	1	4	3	20
1	1	2	3	3
1	6	7	1	4
2	11	10	10	17
6	1	1	5	4

¹ <http://inclass.kaggle.com/>

Example: Heritage Health Length of Stay Data—2

- ▶ Sample mean, standard deviation, and median for the 12,928 claims

(Estimate of μ): $\bar{x} = 4.3$ days

(Estimate of σ): $s = 4.9$ days

(Estimate of population median, m): $\hat{m} = 2$ days

Example: Heritage Health Length of Stay Data—3

Some sample percentiles for the 12,928 claims

Percentile	Sample estimate
2.5 th	1 day
25 th	1 day
50 th	2 days
75 th	5 days
97.5 th	20 days

Summary

- ▶ Summary measures that can be computed on a sample of continuous data include the mean, standard deviation, median (50th percentile), and other percentiles
- ▶ These sample-based estimates are the best estimates of unknown, underlying population quantities. For example:
 - ▶ $\bar{x}$ is the best estimate of the population mean (μ)
 - ▶ s is the best estimate of the population standard deviation (σ)
- ▶ (Soon) we will discuss how to address the uncertainty in the estimates of certain sample quantities (ex: mean)



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Continuous Data: Visual Displays

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Learning Objectives

- ▶ Upon completion of this lecture section, you will be able to:
 - ▶ Utilize histograms and boxplots to visualize the distributions of samples of continuous data
 - ▶ Identify key summary statistics on the boxplot
 - ▶ Name and describe basic characteristics of some common distribution shapes for continuous data

Pictures of Data: Continuous Variables

- ▶ Histograms and boxplots
 - ▶ Means, standard deviations, and percentile values do not tell the whole story of data distributions
 - ▶ Differences in shape of the distribution
 - ▶ Histograms are a way of displaying the distribution of a set of data by charting the number (or percentage) of observations whose values fall within pre-defined numerical ranges
 - ▶ Boxplots are graphics that display key characteristics of a dataset: these are especially nice tools for comparing data from multiple samples visually

Example: Histogram, 113 Systolic Blood Pressures

- ▶ Data on systolic blood pressures (SBP) from a random clinical sample of 113 men
- ▶ A histogram can be created by:
 - ▶ Breaking the data (blood pressure) range into bins of equal width
 - ▶ Counting the number of the 113 observations whose blood pressure values fall within each bin
 - ▶ Plotting the number (or relative frequency) of observations that fall within each bin as a bar graph

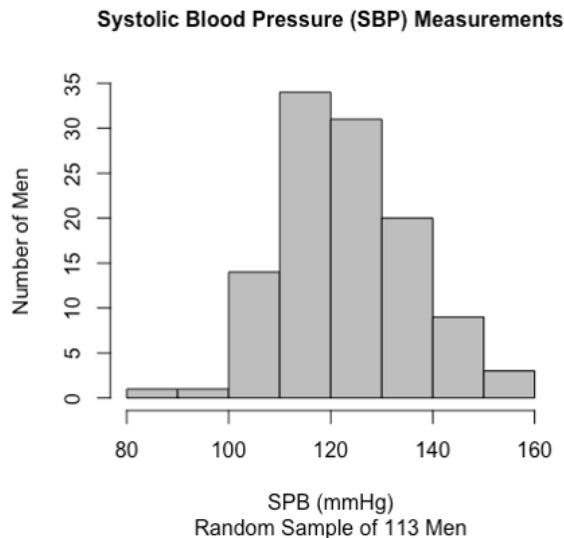
Example: Histogram, 113 Systolic Blood Pressures: Number of Observations

- ▶ A basic histogram of these 113 measurements
 - ▶ Number of observations on the vertical axis

$$\bar{x} = 123.6 \text{ mmHg}$$

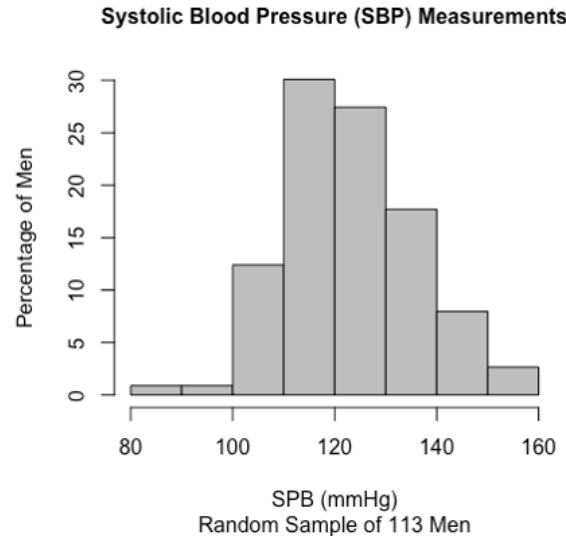
$$s = 12.9 \text{ mmHg}$$

$$\hat{m} = 123.0 \text{ mmHg}$$



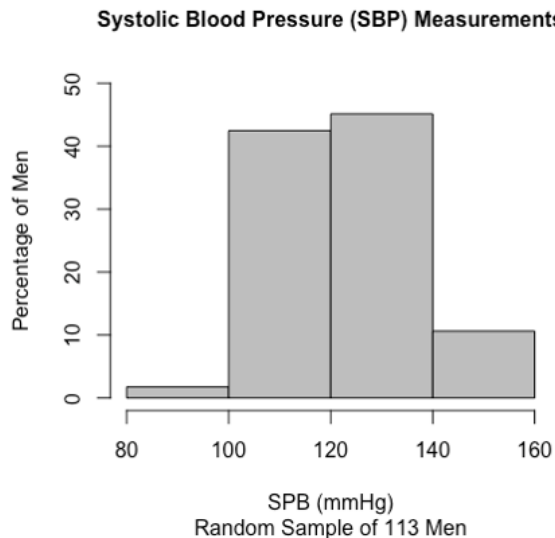
Example: Histogram, 113 Systolic Blood Pressures: Percentage of Observations

- ▶ A basic histogram of these 113 measurements
 - ▶ Percentage of observations on the vertical axis



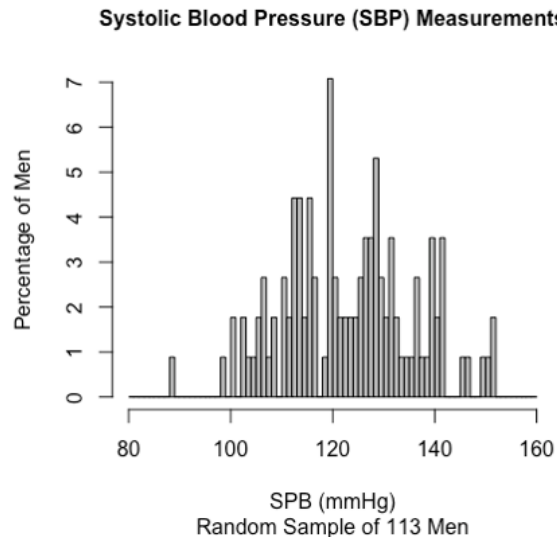
Example: Histogram, 113 Systolic Blood Pressures: Larger Bin Width

- ▶ A basic histogram of these 113 measurements
 - ▶ Percentage of observations on the vertical axis, larger bin width



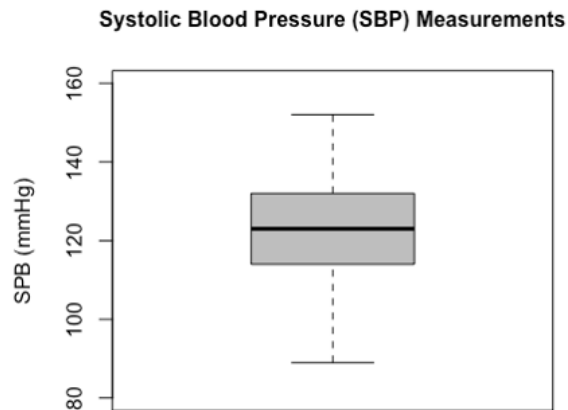
Example: Histogram, 113 Systolic Blood Pressures: Narrower Bin Width

- ▶ A basic histogram of these 113 measurements
 - ▶ Percentage of observations on the vertical axis, narrower bin width



Example: Boxplot, 113 Systolic Blood Pressures

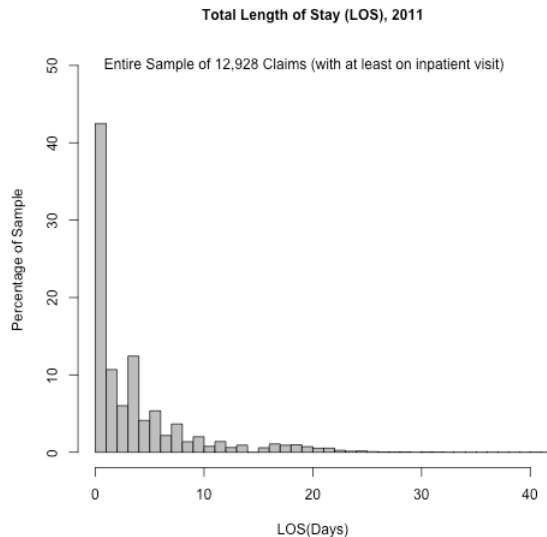
- ▶ A boxplot of the same 113 SBP measurements



Example: Histogram, Heritage Health Length of Stay Data—1

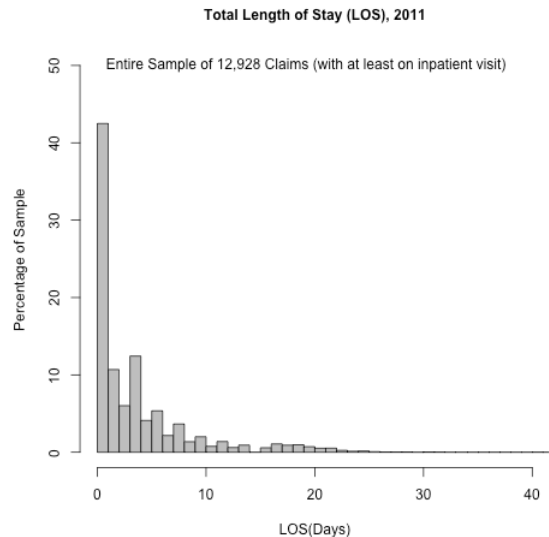
► Histogram of the 12,928 length of stay values

$$\begin{aligned}\bar{x} &= 4.4 \text{ days} \\ s &= 4.7 \text{ days} \\ \hat{m} &= 2.0 \text{ days}\end{aligned}$$



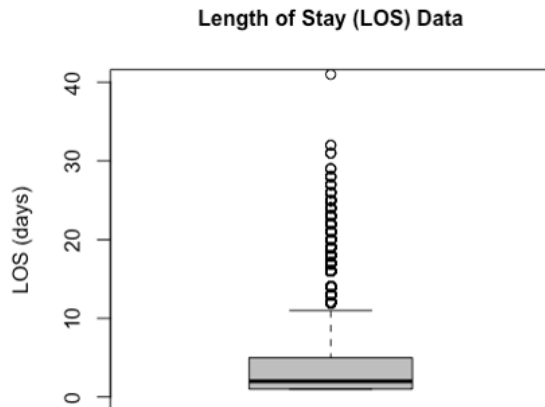
Example: Histogram, Heritage Health Length of Stay Data—2

► How to characterize this distribution?



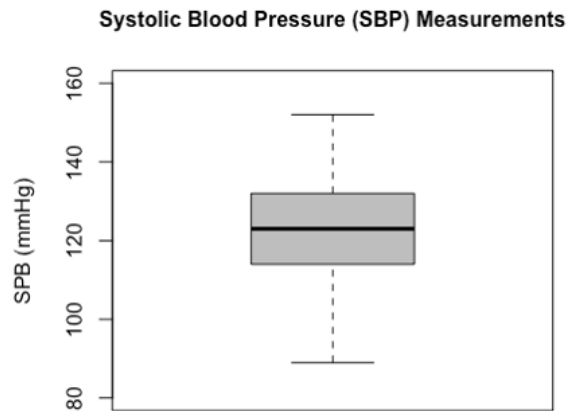
Example: Boxplot, Heritage Health Length of Stay Data

- Boxplot of the 12,928 length of stay values



The “Anatomy” of a Boxplot—2

- ▶ A boxplot of 113 SBP measurements taken from 113 men



Outlier Cutoffs for a Boxplot

- ▶ The cutoff criteria:
 - ▶ “Large” outliers: values $> \text{upper hinge} + 1.5 \cdot \text{IQR}$
i.e. $> 75\text{th percentile} + 1.5 \cdot (75\text{th percentile} - 25\text{th percentile})$
 - ▶ “Small” outliers: values $< \text{lower hinge} - 1.5 \cdot \text{IQR}$
i.e. $< 25\text{th percentile} - 1.5 \cdot (75\text{th percentile} - 25\text{th percentile})$

Some “Common” Distribution Shapes—1

- ▶ Symmetric and “bell-shaped”
- ▶ Right (positively) skewed

Some “Common” Distribution Shapes—2

- ▶ Left (negatively) skewed
- ▶ Uniform

Summary

- ▶ Histograms and boxplots are useful visual tools for characterizing the shape of a data distribution above and beyond the information given by summary statistics
- ▶ Relatively common shapes for samples of continuous data measures include symmetric and “bell” shaped, right skewed, left skewed, and uniform



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Sample Estimate Values, Distributions, and Sample Sizes

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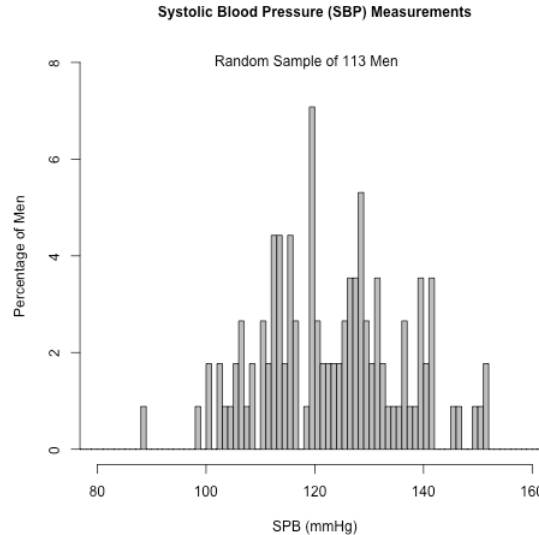
Learning Objectives

- ▶ Upon completion of this lecture section, you will be able to:
 - ▶ Understand that a random sample taken from a larger population will (imperfectly) mimic the characteristics of the larger population
 - ▶ Understand that the distribution of values in a random sample should reflect the distribution of the values in the population from which the sample was taken
 - ▶ Understand and explain that increasing sample size does not systematically decrease the value of sample summary statistic estimates
 - ▶ Begin to understand that while increasing sample size does not decrease sample summary statistic estimates, the estimates become less variable with larger samples

Example: Increasing Sample Size and Sample Distribution: SBP Original Data

- ▶ Histogram of original data, systolic blood pressures (SBP) from 113 men

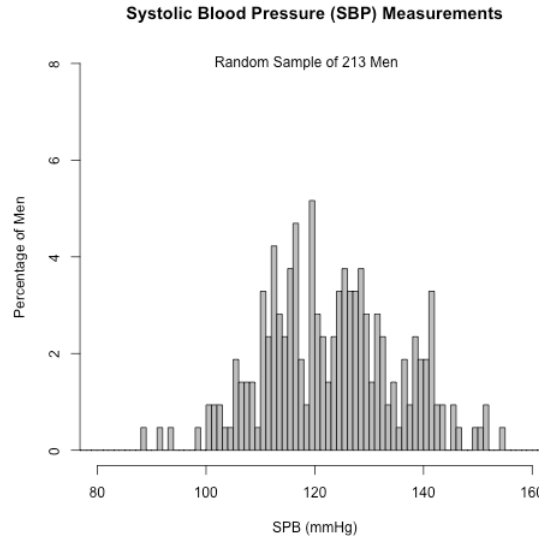
$$\begin{aligned}\bar{x} &= 123.6 \text{ mmHg} \\ s &= 12.9 \text{ mmHg} \\ \hat{m} &= 123.0 \text{ mmHg}\end{aligned}$$



Example: Increasing Sample Size and Sample Distribution: 213 SBP Measurements

- ▶ Histogram of 213 SBP measurements
 - ▶ Original 113 men, plus 100 more randomly selected from same clinical population

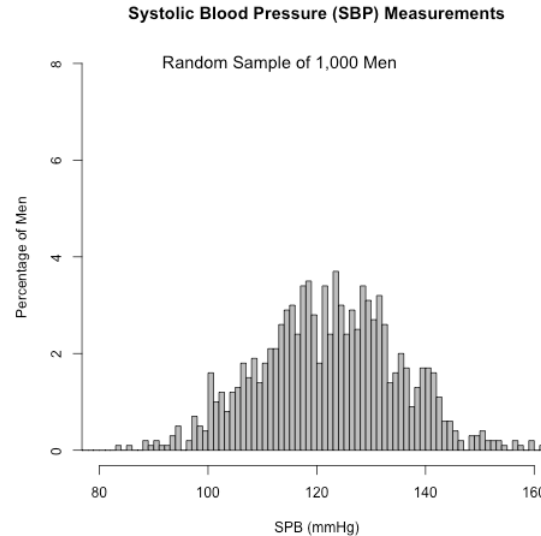
$$\begin{aligned}\bar{x} &= 123.0 \text{ mmHg} \\ s &= 13.0 \text{ mmHg} \\ \hat{m} &= 123.0 \text{ mmHg}\end{aligned}$$



Example: Increasing Sample Size and Sample Distribution: 1,000 SBP Measurements

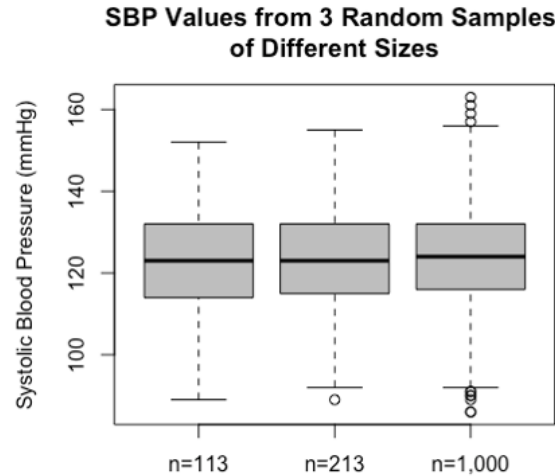
- ▶ Histogram of 1,000 SBP measurements
 - ▶ Original 113 men, plus 887 more randomly selected from same clinical population

$$\begin{aligned}\bar{x} &= 123.2 \text{ mmHg} \\ s &= 13.3 \text{ mmHg} \\ \hat{m} &= 123.0 \text{ mmHg}\end{aligned}$$



Example: Increasing Sample Size and Sample Distribution: SBP Boxplots

- ▶ Boxplots of the individual sample SBP values from the three random samples



Example: Increasing Sample Size and Sample Distribution: SBP Sample Means

Sample means from 5 random samples of men, of sizes $n=113$, 213, and 1,000, respectively

Sample	$n=113$	$n=213$	$n=1,000$
1	124.4	123.5	122.6
2	121.7	123.4	123.6
3	122.6	123.1	122.6
4	123.1	123.4	122.8
5	120.5	123.1	123.1

Example: Increasing Sample Size and Sample Distribution: SBP Sample Standard Deviations

Sample standard deviations from 5 random samples of men, of sizes $n=113$, 213, and 1,000, respectively

Sample	$n=113$	$n=213$	$n=1,000$
1	12.9	13.1	12.4
2	14.4	11.6	13.3
3	12.0	13.1	13.1
4	14.5	12.9	12.9
5	13.6	13.6	13.1

Example: Increasing Sample Size and Sample Distribution: LOS Data

- ▶ Length of stay (LOS) claims at Heritage Health, with an inpatient stay of at least one day in 2011 (12,928 claims)

(Estimate of μ): $\bar{x} = 4.3$ days

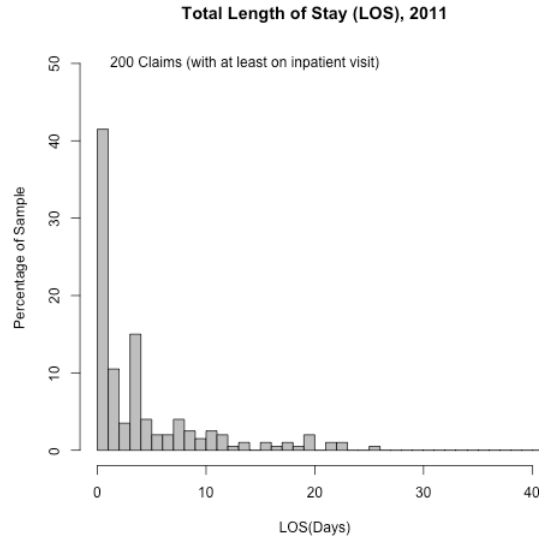
(Estimate of σ): $s = 4.9$ days

(Estimate of population median): $\hat{m} = 2$ days

Example: Increasing Sample Size and Sample Distribution: LOS Data for 200 Patients

- ▶ Random sample of 200 patients, Heritage Health data

$$\begin{aligned}\bar{x} &= 3.8 \text{ days} \\ s &= 4.2 \text{ days} \\ \hat{m} &= 2.0 \text{ days}\end{aligned}$$



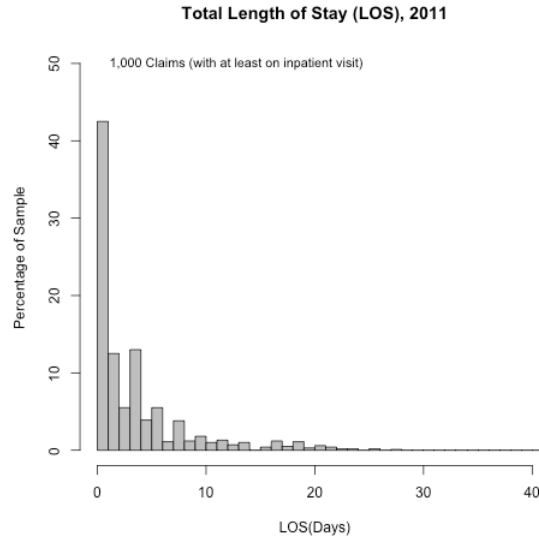
Example: Increasing Sample Size and Sample Distribution: LOS Data for 1,000 Patients

- ▶ Random sample of 1,000 patients, Heritage Health data

$$\bar{x} = 4.3 \text{ days}$$

$$s = 4.7 \text{ days}$$

$$\hat{m} = 2.0 \text{ days}$$



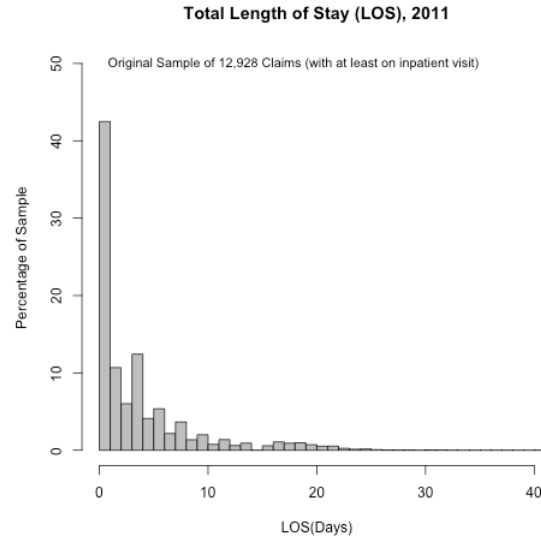
Example: Increasing Sample Size and Sample Distribution: LOS Data for Original Sample

- Original sample of 12,928 patients, Heritage Health data

$$\bar{x} = 4.3 \text{ days}$$

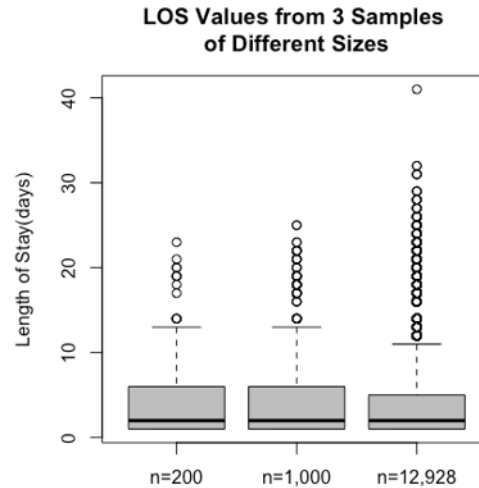
$$s = 4.9 \text{ days}$$

$$\hat{m} = 2.0 \text{ days}$$



Example: Increasing Sample Size and Sample Distribution: LOS Data Boxplot

- Boxplots of the individual sample LOS values from the three samples



Example: Increasing Sample Size and Sample Distribution: LOS Data Sample Means

Sample means from 5 random samples of patients, of sizes $n=200$, 500, and 1,000, respectively

Sample	$n=200$	$n=500$	$n=1,000$
1	4.05	4.19	4.15
2	4.97	4.21	4.12
3	4.06	4.31	4.28
4	4.03	3.99	4.14
5	4.96	4.16	4.24

Example: Increasing Sample Size and Sample Distribution: LOS Data Sample Standard Deviations

Sample standard deviations from 5 random samples of patients, of sizes $n=200$, 500, and 1,000, respectively

Sample	$n=200$	$n=500$	$n=1,000$
1	4.68	4.78	5.02
2	5.67	4.84	4.70
3	4.90	5.10	5.10
4	4.50	4.78	4.74
5	4.45	4.65	4.73

Summary

- ▶ The distribution of sample values of continuous data should (imperfectly) mimic the distribution of the values in the population from which the sample was taken
- ▶ With regard to the distribution of sample values and increasing sample size:
 - ▶ Will not systematically alter the shape of the sample distribution
 - ▶ Will result in a more “filled out” distribution
 - ▶ Will not systematically alter the values of the sample statistic
 - The sample statistic estimates will vary from random sample to random sample but will not systematically get larger (or smaller) with increasing sample size
 - ▶ Will increase the precision of the summary statistics as estimates of the unknown (population level) true values (more to come shortly)



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Comparing Distributions of Continuous Data

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Learning Objectives

- ▶ Upon completion of this lecture section, you will be able to:
 - ▶ Suggest graphical approaches to comparing distributions of continuous data between two or more samples
 - ▶ Explain why a difference in sample means can be used to quantify, in a single number summary, differences in distributions of continuous data

Motivation: Comparisons

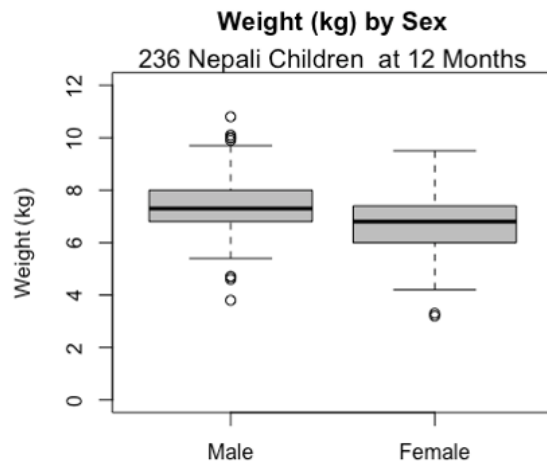
- ▶ Frequently, in public health/medicine/science, etc., researchers/practitioners are interested in comparing two (or more) populations via data collected on samples from these populations
- ▶ Such comparisons can be used to investigate questions, such as:
 - ▶ How does weight change differ between those who are on a low-fat diet compared to those on a low-carbohydrate diet?
 - ▶ How do salaries differ between males and females?
 - ▶ How do cholesterol levels differ across weight groups?
- ▶ While these comparisons can be done visually, it is also useful to have a numerical summary

Motivation: Numerical Summary

- ▶ Theoretically, this numerical summary could be many things:
 - ▶ Difference in medians
 - ▶ Ratio of means
 - ▶ Difference in 95th percentiles
 - ▶ Ratio of standard deviations
 - ▶ Etc.
- ▶ However, what is commonly used (for reasons that we will elaborate on shortly) is a difference in sample means
 - ▶ When comparing sample distributions, this can be a reasonable measure of the overall differences in these distributions (as an estimate of the underlying difference in the population distributions)

Example: Weight Distributions by Sex: Boxplot

- ▶ 236 Nepali children at 12 months old (124 males, 112 females)



Example: Weight Distributions by Sex: Histograms

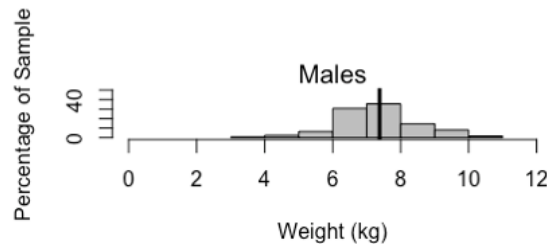
- ▶ 236 Nepali children at 12 months old (124 males, 112 females)



Example: Weight Distributions by Sex: Means

- ▶ 236 Nepali children at 12 months old (124 males, 112 females)
 - ▶ Means indicated by the vertical lines

$$\bar{x}_{males} = 7.4 \text{ kg}$$
$$\bar{x}_{females} = 6.7 \text{ kg}$$



Example: Weight Distributions by Sex: Difference in Means—1

- ▶ Common numerical comparison: difference in means

$$\bar{x}_{male} - \bar{x}_{female} = 7.4 \text{ kg} - 6.7 \text{ kg} = 0.7 \text{ kg}$$

- ▶ On average, male children weigh more than female children by 0.7 kg

Example: Weight Distributions by Sex: Difference in Means—2

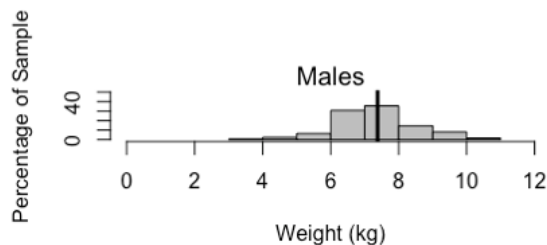
- ▶ Common numerical comparison: difference in means
- ▶ Direction of comparison is arbitrary but important to note!

$$\bar{x}_{female} - \bar{x}_{male} = 6.7 \text{ kg} - 7.4 \text{ kg} = -0.7 \text{ kg}$$

- ▶ On average, female children weigh less than male children by 0.7 kg, which is the same as stating, “on average, male children weigh more than female children by 0.7 kg”

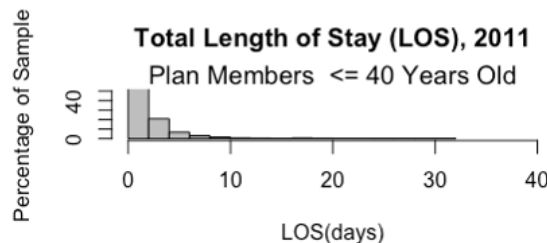
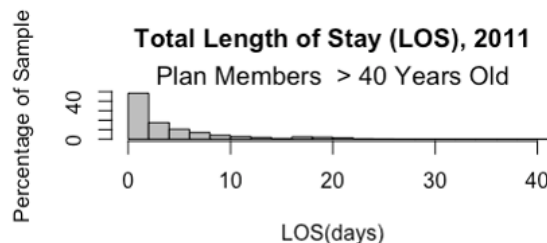
Example: Weight Distributions by Sex: Difference in Means—3

- ▶ Difference in means
 - ▶ Why does this characterize difference in distributions?



Example: Length of Stay by Age at First Claim: Histograms

- ▶ Histograms
 - ▶ 3,769 $\leq$ to 40 years old at first claim in 2011
 - ▶ 9,159 $>$ 40 years at first claim in 2011



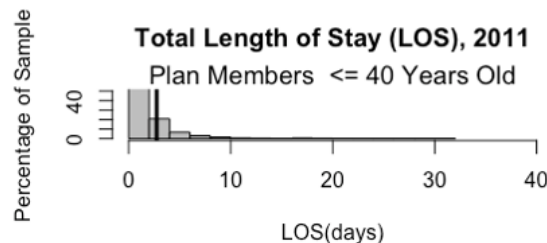
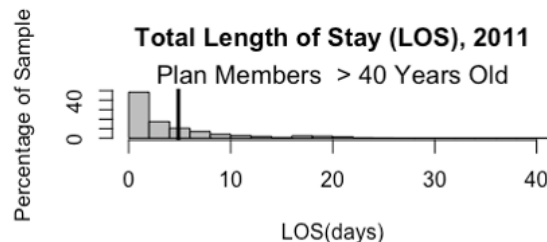
Example: Length of Stay by Age at First Claim: Difference in Means

- ▶ Common numerical comparison: difference in means

$$\bar{x}_{>40 \text{ years}} - \bar{x}_{\leq 40 \text{ years}} = 4.9 \text{ days} - 2.7 \text{ days} = 2.2 \text{ days}$$

Example: Length of Stay by Age at First Claim: Means

► Histograms with vertical lines for means



Example: Academic Physician Salaries

Gender differences in physician salaries, even after adjustment for differences in specialty, institutional characteristics, academic productivity, academic rank, work hours, and other factors

Results The mean salary within our cohort was \$167 669 (95% CI, \$158 417-\$176 922) for women and \$200 433 (95% CI, \$194 249-\$206 617) for men. Male gender was associated with higher salary (+\$13 399; $P=.001$) even after adjustment in the final model for specialty, academic rank, leadership positions, publications, and research time. Peters-Belson analysis (use of coefficients derived from regression model for men applied to women) indicated that the expected mean salary for women, if they retained their other measured characteristics but their gender was male, would be \$12 194 higher than observed.

Conclusion Gender differences in salary exist in this select, homogeneous cohort of mid-career academic physicians, even after adjustment for differences in specialty, institutional characteristics, academic productivity, academic rank, work hours, and other factors.

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Example: Academic Physician Salaries: Bivariable Associations between Salary and Measured Characteristics Table

Table 2. Bivariable Associations Between Salary and Measured Characteristics (continued)

Characteristics	Salary Estimate, \$ (95% CI)	P Value
Current institution National Institutes of Health funding rank group		
First	177 590 (166 584 to 188 596)	<.001
Second	184 771 (173 608 to 195 934)	
Third	188 104 (177 991 to 198 217)	
Fourth	206 621 (196 953 to 216 288)	
Current institution region		
West	194 474 (183 876 to 205 073)	.09
Midwest	198 890 (186 426 to 211 354)	
South	194 439 (182 105 to 206 773)	
Northeast	192 152 (173 878 to 190 425)	
Still at K institution at time of survey		

Example: Academic Physician Salaries: Comparing Means between More than Two Groups

- ▶ General practice for numerically comparing means between more than two groups
 - ▶ Designate one group as the “reference”
 - ▶ Report the differences for the other groups compared to this same reference
- ▶ For example, if we make “West” the reference region, then the mean differences can be reported as:

$$\bar{x}_{MW} - \bar{x}_W = 198,890 - 194,474 = \$4,416$$

$$\bar{x}_S - \bar{x}_W = 194,439 - 194,474 = -\$35$$

$$\bar{x}_{NE} - \bar{x}_W = 192,152 - 194,474 = -\$2,322$$

Example: Academic Physician Salaries: Reference Group

- ▶ No need to change the reference group to get other mean differences

Summary

- ▶ While the distributions of continuous data can be compared between samples in many ways, some key approaches include:
 - ▶ Visual comparisons, such as side-by-side boxplots
 - ▶ Numerical comparisons, mainly the mean difference, between any group of two samples



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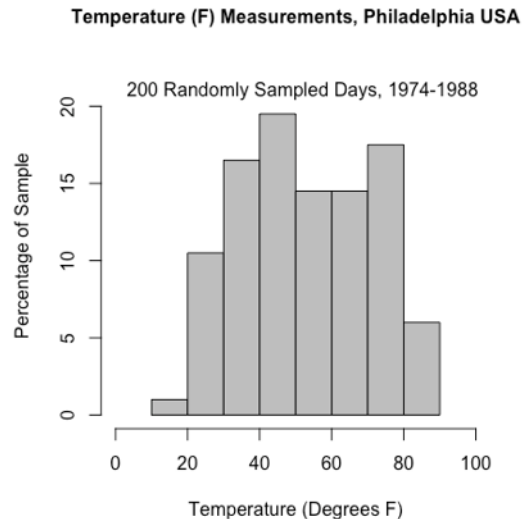
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Additional Examples

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Example: Daily Temperatures in Philadelphia

- ▶ Random sample of 200 days taken from the period 1974–1988



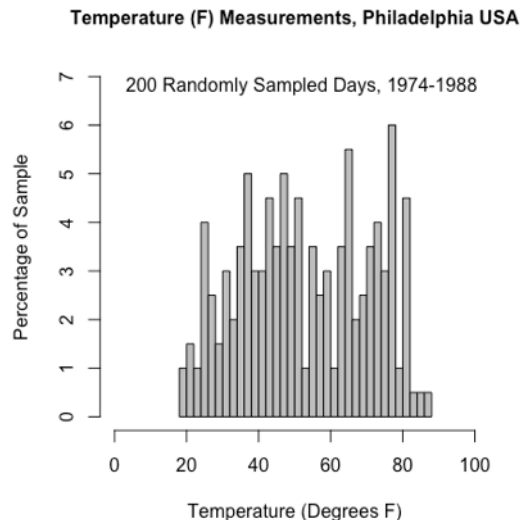
Example: Daily Temperatures in Philadelphia: Narrower Bins

- ▶ Random sample of 200 days taken from the period 1974–1988, with narrower bins

$$\bar{x} = 52.9 \text{ degrees}$$

$$s = 17.9 \text{ degrees}$$

$$\hat{m} = 50.6 \text{ degrees}$$



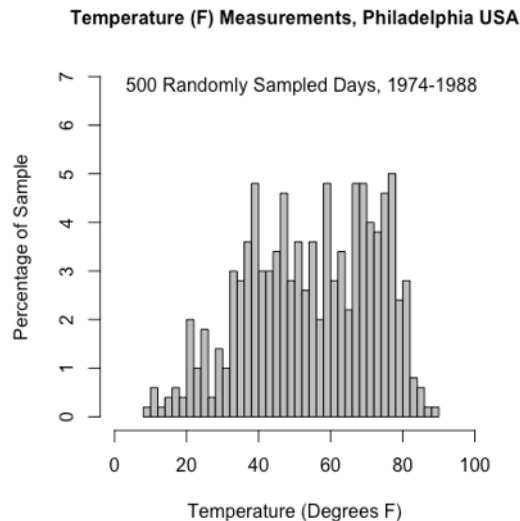
Example: Daily Temperatures in Philadelphia: 500 Days

- ▶ Random sample of 500 days taken from the period 1974–1988

$$\bar{x} = 54.8 \text{ degrees}$$

$$s = 17.7 \text{ degrees}$$

$$\hat{m} = 55.9 \text{ degrees}$$



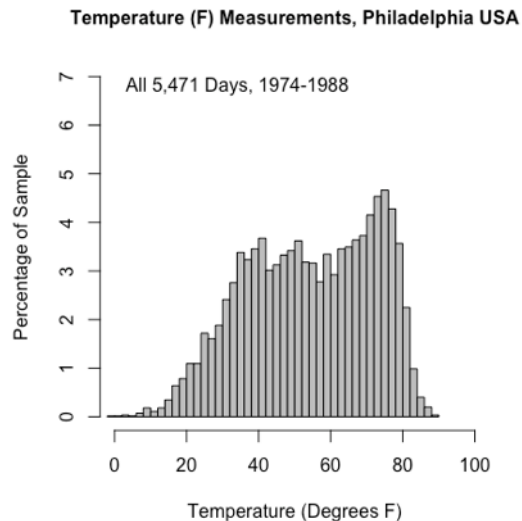
Example: Daily Temperatures in Philadelphia: All Days

- All days in the period 1974–1988

$$\bar{x} = 54.4 \text{ degrees}$$

$$s = 17.9 \text{ degrees}$$

$$\hat{m} = 55.3 \text{ degrees}$$



Example: Daily Temperatures in Philadelphia: Sample Means

Sample means from 5 random samples of days, of sizes $n=200$, 500, and 1,000, respectively

Sample	$n=200$	$n=500$	$n=1,000$
1	56.1	54.3	53.9
2	52.3	53.2	55.0
3	54.0	54.4	54.7
4	52.7	54.3	54.3
5	55.0	53.7	53.7

Example: Daily Temperatures in Philadelphia: Sample Standard Deviations

Sample standard deviations from 5 random samples of patients, of sizes $n=200$, 500 , and $1,000$, respectively

Sample	n=200	n=500	n=1,000
1	18.1	18.1	17.9
2	18.4	17.8	18.0
3	17.8	17.1	17.7
4	17.3	16.9	17.8
5	17.4	18.1	18.0

Example: Daily Temperatures in Philadelphia: Sample Medians

Sample medians from 5 random samples of patients, of sizes $n=200$, 500, and 1,000, respectively

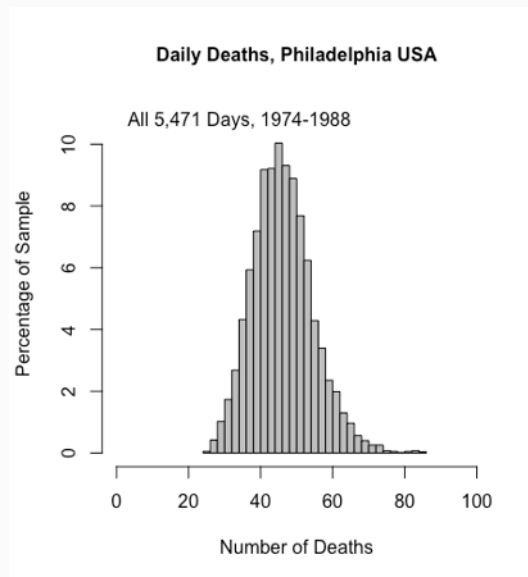
Sample	$n=200$	$n=500$	$n=1,000$
1	56.2	56.5	54.3
2	53.5	52.9	55.5
3	55.6	55.5	56.6
4	51.8	54.0	55.0
5	55.5	54.5	54.1

Example: Daily Death Counts in Philadelphia

$\bar{x} = 46.7$ deaths

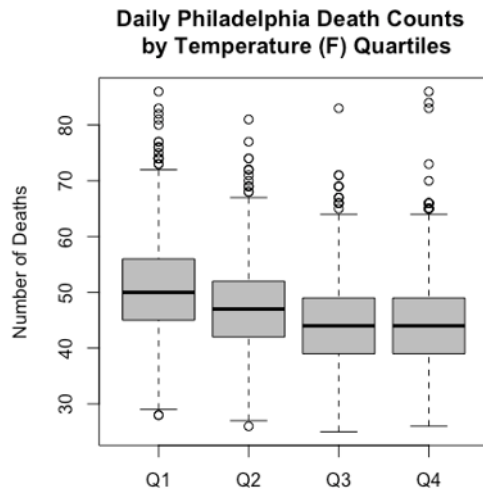
$s = 8.4$ deaths

$\hat{m} = 46.0$ deaths



Example: Daily Death Counts in Philadelphia by Temperature Quartiles

Quartiles: Side-by-side Boxplots



Example: Daily Death Counts in Philadelphia by Temperature

Quartiles: Means

► Mean deaths for each temperature quartile:

$$\bar{x}_{Q_1} = 50.8 \quad (-0.25^\circ\text{F to } 40^\circ\text{F})$$

$$\bar{x}_{Q_2} = 47.4 \quad (40.0^\circ\text{F to } 55.3^\circ\text{F})$$

$$\bar{x}_{Q_3} = 44.3 \quad (55.2^\circ\text{F to } 70.2^\circ\text{F})$$

$$\bar{x}_{Q_4} = 44.2 \quad (70.2^\circ\text{F to } 88.5^\circ\text{F})$$